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Lab 8: Wireshark DHCP

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Introduction

- Investigate Dynamic Host Configuration Protocol (DHCP) behavior.
- Observe DHCP Discover, Offer, Request, and ACK messages.
- Understand dynamic IP address assignment.
- Use Wireshark to capture and analyze DHCP traffic.

Note

DHCP is critical for automating network setup without needing manual IP address assignment.

"Tell me and I forget. Show me and I remember. Involve me and I understand."

– Chinese proverb

What is DHCP?

- **Dynamic Host Configuration Protocol (DHCP):**
  - Assigns IP addresses and network configuration automatically.
  - Used in corporate, university, and home networks.
- **Key Messages:**
  - DHCP Discover, Offer, Request, ACK.
- **Benefits:**
  - Simplifies IP management.
  - Reduces configuration errors.

Note

Each DHCP message plays a specific role in establishing and confirming the client-server communication.

How DHCP Works

- Client broadcasts DHCP **Discover**.
- Server replies with DHCP **Offer**.
- Client requests offered IP via DHCP **Request**.
- Server acknowledges with DHCP **ACK**.

Note

Pay attention to the transaction ID; it is used to match requests and responses.

Capturing Packets

- **Mac/Linux:**
  - Flush IP settings and restart DHCP client.
- **Windows:**
  - Use `ipconfig /release` and `ipconfig /renew`.
- **Use Wireshark filter:**
  - `bootp` or `dhcp`

Note

Filtering using `bootp` will help you quickly isolate DHCP messages from other traffic.

Analyzing DHCP Discover

- **Q1:** What transport protocol is used for the DHCP Discover message: UDP or TCP? *(Check the underlying protocol in the Layer 4 details of the packet.)*
- **Q2:** What is the source IP address used in the IP datagram containing the Discover message? Is there anything special about this address? *(Check the source IP field; explain why it may be 0.0.0.0.)*
- **Q3:** What is the destination IP address used in the datagram containing the Discover message? Is there anything special about this address? *(Look for a broadcast address like 255.255.255.255.)*
- **Q4:** What is the value in the transaction ID field of the DHCP Discover message? *(Locate the transaction ID in the DHCP details.)*
- **Q5:** Inspect the options field in the DHCP Discover message. What are five pieces of information (beyond an IP address) that the client is suggesting or requesting to receive from the DHCP server as part of this DHCP transaction? *(Look for options like DNS, domain name, subnet mask, etc.)*

Analyzing DHCP Offer

- **Q6:** How do you know that the DHCP Offer message is being sent in response to the DHCP Discover message?
  (Match the Transaction ID and possibly the MAC address.)
- **Q7:** What is the source IP address in the IP datagram containing the Offer message? Is there anything special about this address?
  (Check for the DHCP server's IP address.)
- **Q8:** What is the destination IP address used in the datagram containing the Offer message? Is there anything special about this address?
  (Check if the address is a broadcast or specific client IP.)
- **Q9:** What are five pieces of information that the DHCP server is providing to the client in the Offer message?
  (Look for offered IP address, lease time, DNS server, gateway IP.)

Analyzing DHCP Request

- **Q10:** What is the UDP source port number in the IP datagram containing the DHCP Request message? What is the UDP destination port number being used?
  (Look for source 68, destination 67.)
- **Q11:** What is the source IP address in the IP datagram containing this Request message? Is there anything special about this address?
  (If still 0.0.0.0, client hasn't yet received IP.)
- **Q12:** What is the destination IP address used in the datagram containing this Request message? Is there anything special about this address?
  (Usually broadcast 255.255.255.255.)
- **Q13:** What is the value in the transaction ID field of the DHCP Request message? Does it match the transaction IDs of the earlier Discover and Offer messages?
  (Transaction IDs should be identical.)
- **Q14:** What differences do you see between the entries in the 'Parameter Request List' option in the DHCP Request message and the same list option in the earlier Discover message?

Analyzing DHCP ACK

- **Q15:** What is the source IP address in the IP datagram containing the DHCP ACK message? Is there anything special about this address?
  (Check for the server's IP.)
- **Q16:** What is the destination IP address used in the datagram containing this ACK message? Is there anything special about this address?
  (Check if it is the client's or broadcast.)
- **Q17:** What is the name of the field in the DHCP ACK message that contains the assigned client IP address?
  (Look for "Your IP Address" field.)
- **Q18:** For how long has the DHCP server assigned this IP address to the client?
  (Find the lease time in the options field.)
- **Q19:** What is the IP address of the first-hop router on the default path from the client to the rest of the Internet, as provided in the DHCP ACK message?
  (Look for Option 3: Router field.)

Additional Tips

- **Use Filters:**
  - `bootp.option.dhcp == 1` (Discover)
  - `bootp.option.dhcp == 2` (Offer)
  - `bootp.option.dhcp == 3` (Request)
  - `bootp.option.dhcp == 5` (ACK)
- **Match Transaction IDs:** Ensure same session.
- **Broadcast Behavior:** Discover and Request are broadcasts.

Note

Timestamps might not be perfectly synchronized; focus on message sequence and transaction ID.

Conclusion

- DHCP automates network configuration.
- **Observed the full process:** Discover $\rightarrow$ Offer $\rightarrow$ Request $\rightarrow$ ACK.
- **Submission Requirements:**
  - Annotated Wireshark captures.
  - Highlight important packet fields.

Reminder

Always double-check packet details: Transaction ID, Client IP, Lease time, and Gateway IP.